Present Connection

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Technical assignment for a candidate for an internship developer position at the IDT team

Task Description

Build a web application for booking meeting rooms in an office. Users should be able to see the available rooms, book them for meetings and manage their reservations. The system should also show how the rooms are actually used, so the office can see which rooms are often booked but rarely used.

Functional requirements

1. Users page

Display a list of all meeting rooms. Each room should show:

- Name and unique identifier
- Capacity (maximum number of people)
- Floor / location
- Equipment: Projector, TV screen, Video conferencing, Whiteboard (a room can have several)
- Status: active or inactive (inactive rooms cannot be booked, e.g. during renovation)

Filtering: the page must allow filtering rooms by minimum capacity, equipment and floor.

2. Users page

Display a list of all users. Each entry should show:

- First name, last name and unique identifier
- How many active reservations, user has

3. Reservations page

Display a list of reservations with the following information per reservation:

- Room and organizer (user)
- Meeting title
- Start and end date
- Number of attendees
- Status – one of: Booked, Completed, Cancelled

Actions: create a reservation, cancel a reservation, end a meeting early.

Filtering: the page must allow filtering reservations by room, organizer, status and date range.

4. Booking rules

Every new or edited reservation must be validated in the back end against the rules below. If a rule is broken, the reservation is rejected and the API returns a clear error code and a human-readable message, which the front end displays to the user.

#	Rule	Details
R1	**Maximum duration**	A single reservation cannot last longer than 7 days. Exactly 7 days is allowed.
R2	**No overlapping room bookings**	A room cannot have two active reservations. Cancelled reservation does not block the room.
R3	**No overlapping organizer bookings**	The same user cannot organize two meetings that overlap in date, even in different rooms.
R4	**Valid time range**	Start must in the future.
R5	**Capacity**	Number of attendees must not exceed the room capacity.
R6	**Booking horizon**	A reservation cannot start more than 90 days in the future.
R7	**Active rooms only**	Inactive rooms cannot be booked.

All limits (7 days, 90 days) must be **configurable** (e.g. in appsettings.json), not hard-coded.

Technical requirements

- The **back end** should be built using ASP.NET and an EF Core in-memory database to store information
- The **front end** should be built using React and can use a modern UI library such as Bootstrap, Material UI, Tailwind, etc.
- All calculation logic must be implemented in the back-end
- Code must be shared via GitHub
- No authentication and authorization is needed

Nice to have (optional)

- Unit tests in the backend
- Dependency injection
- Mapper package or custom solution
- TypeScript
- Database seeders
- Calendar / timeline view of a room’s day or week
- Export of the room usage report to PDF

Evaluation criteria

- Functionality of the web app (could not be 100% finished)

- Clean code
- Decisions on architecture for both the back end and front end
- Tools used to build this app
- Design patterns